course goal: gate algorithm

2025, 26, 27

In depth

51, 52, 53

Lec_54

Max and Min Heapify

heap tree

ACBT

parent

The process in which the binary tree is reshaped into a Heap data structure

is called heapify

- Means ,Insertion and Deletion in heap tree is called Heapify
- For max Heap Insertion and deletion ,max heapify
- For min heap Insertion and deletion ,min heapify

max Heapify ⇒ In, de max heap

min heap ⇒ In, de min heap

Heapify → Insertion & deletion

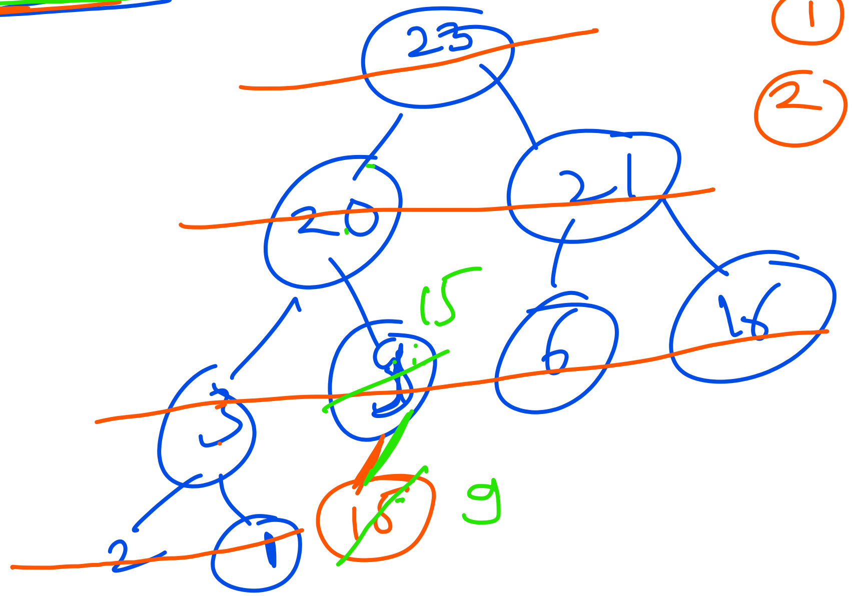
Insertion in Heap : In Any heap insertion Always Happens at first Empty Slot ,Than We place it at Correct place

1) ACBT
2) Parent > child

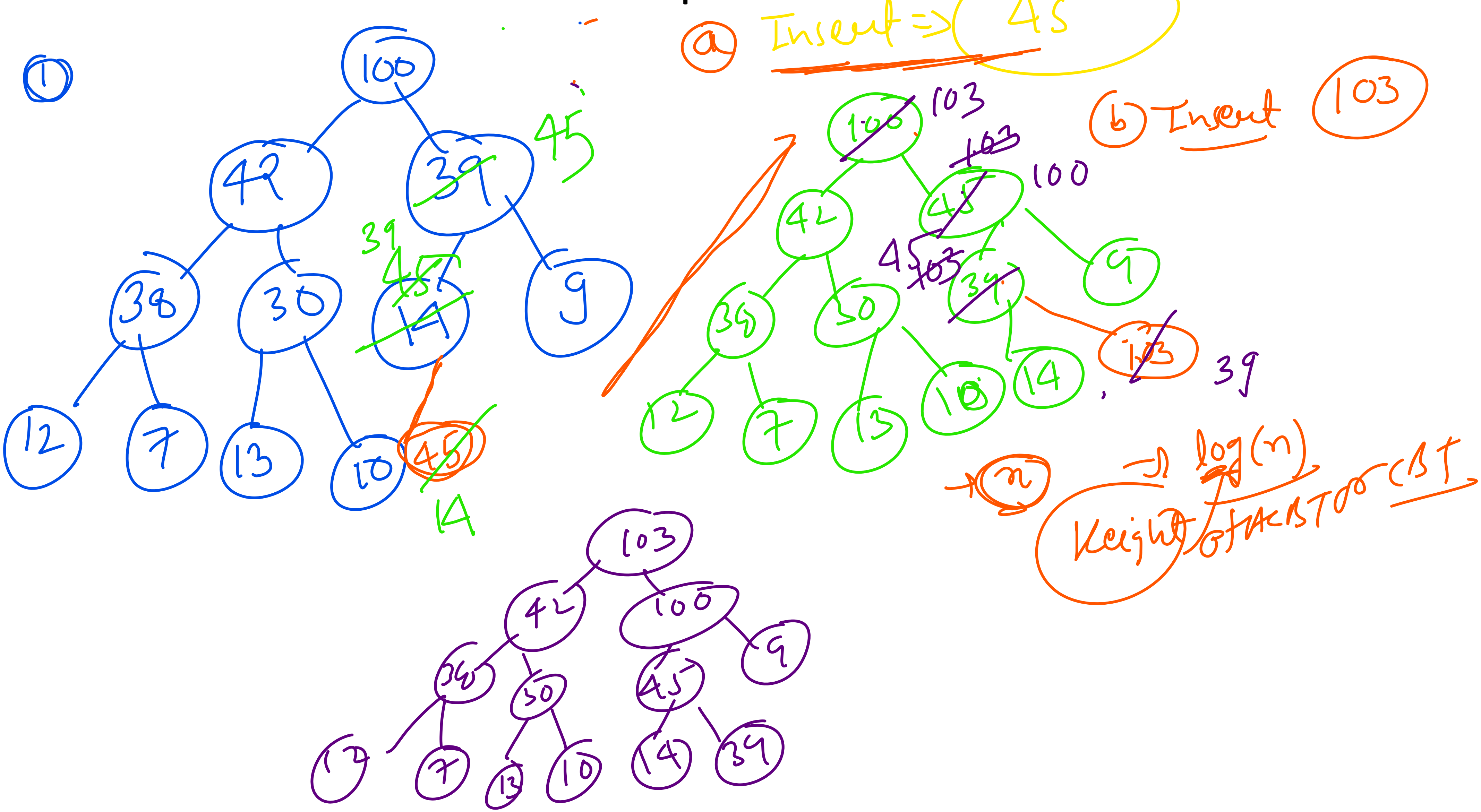
let

Ex: Example of maxHeap

Insert ⇒ 15



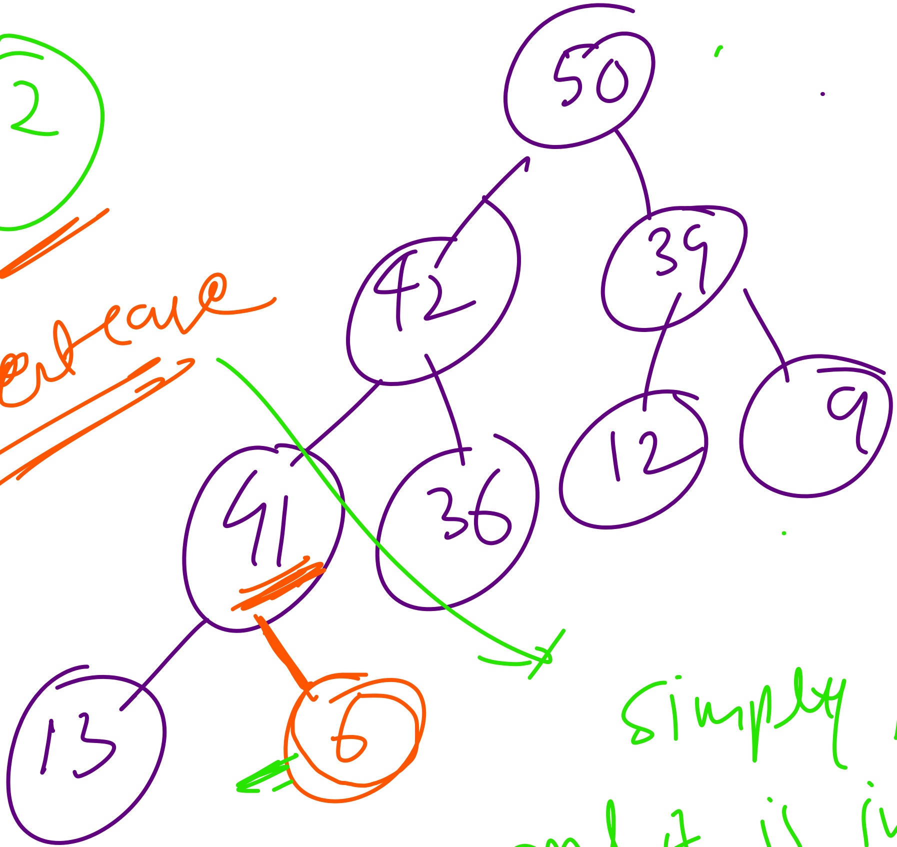
Insertion In Max Heap:



Insert ⇒ 6

2

Best case



Simply insert and it is in correct place

TC and space complexity

space = constant

TC = ? ⇒ WC, AC = ? $O(\log n)$

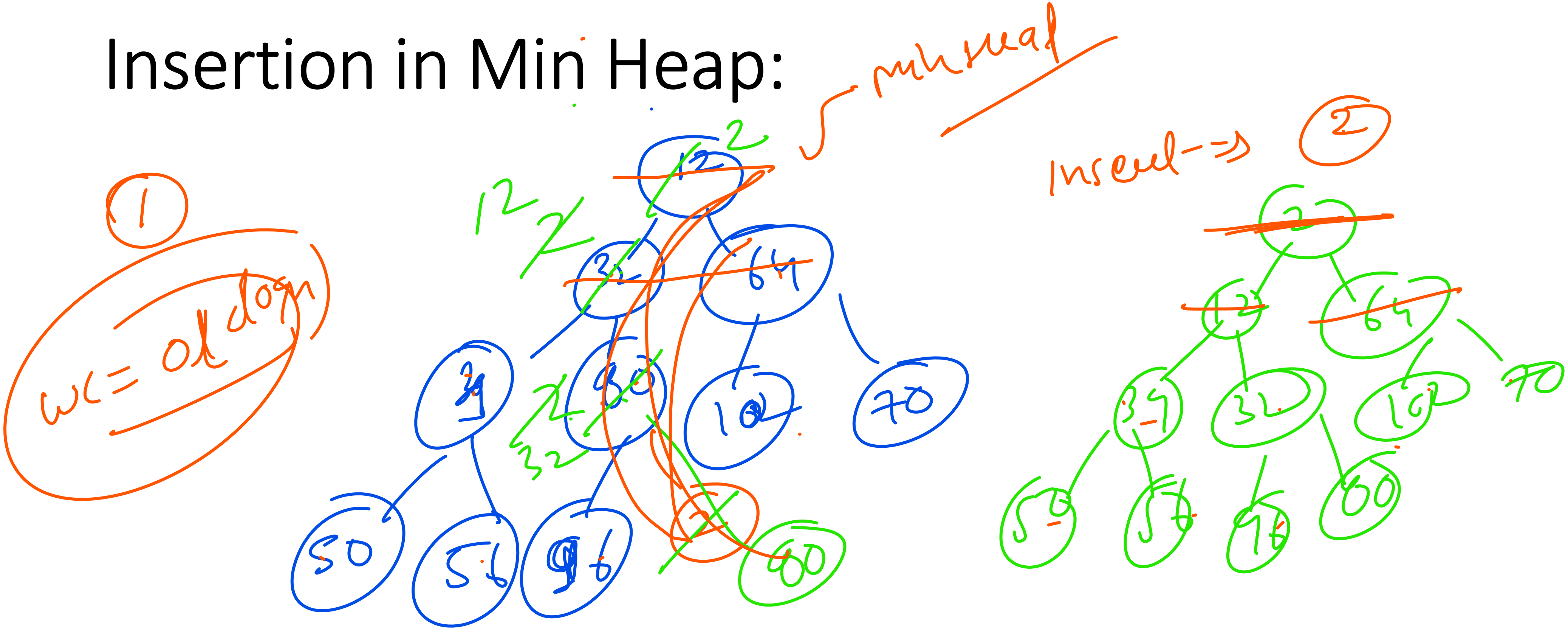
BC = ? $O(1)$

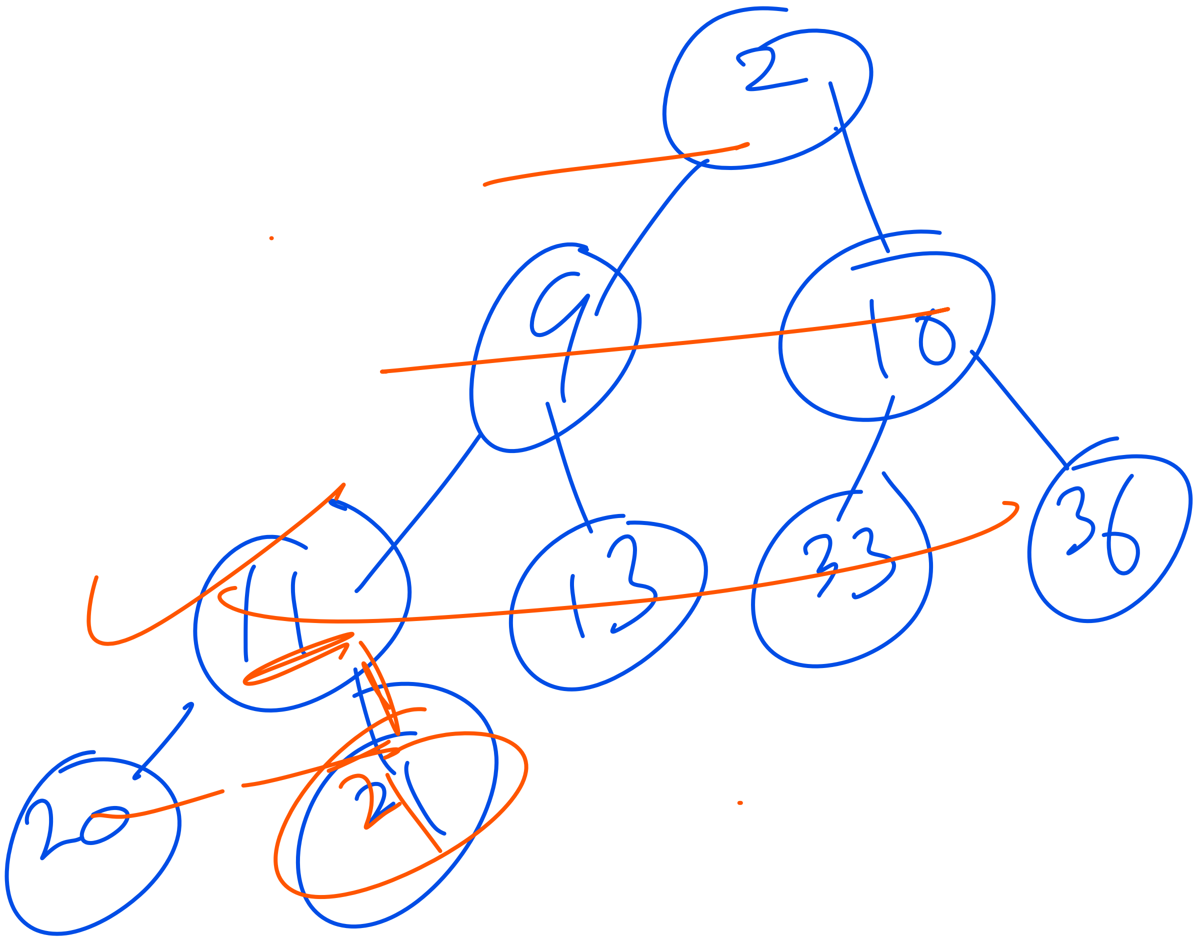
→ In worst case you need to visit first empty slot to root (height of tree)

For Insertion in map

$$TC \begin{cases} BC = O(1) \\ AC, WC = O(\log n) \end{cases}$$

Insertion in Min Heap:





Insert-21

$BC = O(1)$

$WC = AC = O(\log n)$

Time and Space Complexity of Insertion in Heap:

$BC = O(1)$

$WC = AC = O(\log n)$

for both heap

Space $= O(1)$

SS